CRAP and the four moves every designer makes

Week 2, Lecture 1, Design for Builders

Autumn 2026

What we'll cover

- What CRAP stands for and why the acronym is deliberate
- Contrast: the primary tool for directing attention
- Repetition: how recurrence signals a system
- Alignment: the invisible grid that makes layouts feel intentional
- Proximity: whitespace as a grouping tool
- Why beginners consistently skip alignment and how to stop

Context: why visual principles matter before color or typefaces

Your brain judges before you think

Every interface communicates before the user reads a single word.

- Visual weight signals importance: the eye goes to what is heaviest first
- Spacing signals relationship: close elements feel related, distant ones feel separate
- Consistency signals system: repeated patterns feel professional, arbitrary variation feels cheap
- These are not style preferences, they are perceptual facts

Gordon (NN/G, 2020): "Scale, visual hierarchy, balance, contrast, and Gestalt are the five visual design principles most directly connected to usability."

CRAP in one sentence

Robin Williams (2004) named four principles that distinguish a designed layout from a random one: **Contrast**, **Repetition**, **Alignment**, and **Proximity**.

- Every intentional design decision maps to at least one of these four
- Most beginner mistakes break one of them unintentionally
- You can audit any layout by asking: "Which of the four is violated here?"
- The acronym is memorable because it makes you hesitate before breaking a rule

Contrast

Contrast: the primary attention signal

Contrast is the difference between elements that makes one more visually prominent than another.

- **Size contrast:** a 32px heading vs. a 14px body creates a 2x hierarchy level in one move
- **Weight contrast:** bold vs. regular communicates primary vs. secondary within the same size
- **Color contrast:** a saturated accent on a neutral field pulls the eye instantly
- **Space contrast:** an element surrounded by whitespace reads as more important than one packed in

Gordon (2020): "Contrast helps you guide users to key content and actions. Without it, every element competes equally, meaning nothing is emphasized."

What low contrast looks like

Every text element is the same size and weight. The page title, subheading, body copy, and label all render at 14px, color #555.

After (contrast applied)

The eye now knows where to start.

Before (low contrast)

The eye has no entry point.

Page title: 32px, weight 700, color #111.
Subheading: 18px, weight 600, color #333.
Body: 14px, weight 400, color #555.
Label: 12px, weight 500, color #888, uppercase.

Contrast and visual weight

Visual weight is how much visual mass an element has in the composition.

- Large, dark, dense, or isolated elements read as heavier
- Light, small, thin, or grouped elements read as lighter
- Weight is relative: a 24px heading only reads as important if the body copy is smaller
- Malewicz (2022): "Beginners make everything the same size. That's the single most common mistake."

Rule of thumb: no two adjacent text elements should be the same size *and* the same weight.

Repetition

Repetition: recurrence signals system

Repetition means using the same visual treatment every time the same kind of element appears.

- A button style used consistently reads as "button" on first encounter
- A card style used consistently reads as "item in a list" without explanation
- A heading style used consistently tells the eye how deep into the hierarchy it is
- Broken repetition (one card with a drop shadow, the others without) reads as error or exception

Wathan and Schoger (2018): "Every design decision you make becomes part of a design language. Repetition is how you make that language readable."

What broken repetition costs you

When repetition breaks, the user has to decide: is this difference intentional or accidental?

- Intentional difference: "this card is featured" or "this button is destructive"
- Accidental difference: a padding value changed during development, or a font loaded late

The problem: users cannot tell the difference. Both read as information. Accidental inconsistency teaches users to distrust your interface.

Audit your components: count how many distinct button treatments exist in your product. The correct number is two or three. More than five is repetition breakdown.

Alignment

Alignment: the invisible grid

Every element on a screen has an implied relationship with the elements around it.

- Aligned elements share an edge, center, or baseline with at least one other element
- Unaligned elements float: they belong to neither the element above them nor the element below
- The grid does not have to be visible, the reader infers it from consistent placement
- A misaligned label, icon, or heading interrupts the inferred grid and reads as an error

Gordon (2020): "When elements are aligned on a page, the result is a stronger cohesive unit."

Why beginners ignore alignment

Alignment failures are invisible to the untrained eye, until they are pointed out.

- "It looks fine to me" is the most common defense of misaligned layouts
- The reason: the brain corrects small offsets automatically during casual browsing
- But "fine" is not "trustworthy": accumulated small misalignments produce a layout that feels slightly off without the viewer being able to name why
- The fix: grid overlay. If you cannot explain which grid line each element snaps to, it is probably floating.

Try this: in Figma, turn on the layout grid after placing your elements. How many need to move?

Alignment in practice

- Icon and label at different vertical centers
- Heading left-aligned, button center-aligned, on the same card
- Two cards side by side with different top padding
- Text block that starts 12px left of the image above it

Check every element against one anchor: the left edge of the content column, a shared baseline, or the auto-layout padding. Pick one and apply it consistently.

Common alignment mistakes

The fix each time

Proximity

Proximity: distance encodes relationship

Elements placed close together are perceived as belonging to the same group. Elements placed far apart are perceived as separate.

- This is not a convention the designer teaches, the brain applies it automatically
- It is the primary tool for creating visual sections within a layout without drawing lines
- The gap between two groups must be meaningfully larger than the gap within a group
- If inner and outer spacing are equal, the layout reads as one undifferentiated block

NN/G Gestalt video (2020): proximity is the strongest Gestalt grouping signal because it requires no shared visual property, only position.

Proximity in real interfaces

A form with three fields for shipping address and two fields for billing address reads as one group if the spacing is equal throughout.

Add 24px between the two groups and 8px between fields within each group: the form now reads as two distinct tasks without a single label or heading being added.

The principle: **inner gap smaller than outer gap**, always.

- Inner gap (between related items): 4-8px
- Outer gap (between unrelated sections): 24-48px
- The ratio of outer-to-inner gap must be visible at a glance, not measurable with a ruler

Putting it together

CRAP as a checklist, not a style

When you finish any layout, run the four-question audit:

1. **Contrast:** can you name the first, second, and third element the eye should reach? Are they in the right order?
2. **Repetition:** is every recurring element type treated identically? Find every exception.
3. **Alignment:** does every element share an edge or center with at least one other element? Check with grid overlay.
4. **Proximity:** is the whitespace between unrelated groups larger than the whitespace within groups? Measure the ratio.

If any answer is no, you have found the next design fix.

The CRAP audit: worked example

Audit a hypothetical settings card (label + toggle + description):

- **Contrast:** toggle active state has higher contrast than inactive. Label is bolder than description. Pass.
- **Repetition:** three cards in the list. All three have identical padding, identical label size, identical toggle position. Pass.
- **Alignment:** label, toggle, and description all left-aligned. Toggle right-aligned on the same baseline as the label. Pass.
- **Proximity:** label and toggle read as one unit (8px gap). Description reads as explanatory text (16px below label). Outer gap between cards is 24px. Pass.

Take-aways

1. Contrast directs attention. If nothing has high contrast, nothing is emphasized, and the layout fails to communicate hierarchy.
2. Repetition builds trust. Consistent treatment of recurring elements tells the user what kind of thing they are looking at before they read it.
3. Alignment creates cohesion. Every floating element interrupts an inferred grid the reader was relying on.
4. Proximity groups without words. The ratio of inner gap to outer gap communicates relationship before the user reads any label.
5. CRAP is a diagnostic, not a style: use it to find violations, not to produce aesthetics.

> "When nothing is emphasized, nothing is emphasized."
>
> Adam Wathan and Steve Schoger, Refactoring UI, 2018

What's next

- **Section this week:** Design critique. Bring three screenshots of real interfaces. You will annotate and present the visual hierarchy of one of them.
- **Week 2 reading:** Visual literacy: CRAP, Gestalt, and visual hierarchy. Read before section. Sections 1-4 cover today's content; sections 5-8 cover Lecture 2.
- **Lecture 2 this week:** Gestalt: proximity grouping, similarity grouping, closure, and figure-ground.
- **HW1 due this week:** Your 12-component Figma starter kit. Run a CRAP audit on it before you submit.

References

- Gordon, Kelley. "5 Principles of Visual Design in UX." Nielsen Norman Group, 2020. nngroup.com/articles/principles-visual-design/
- Malewicz, Michal. "Master Visual Hierarchy: Principles of Visual Design." YouTube, 2022. youtube.com/watch?v=LfkDiDyn6nU
- Wathan, Adam, and Steve Schoger. "7 Practical Tips for Cheating at Design." Refactoring UI, 2018. medium.com/refactoring-ui/7-practical-tips-for-cheating-at-design-40c736799886
- Wathan, Adam, and Steve Schoger. Refactoring UI. 2018. refactoringui.com